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ProbeSpeak: Last-Mile Fault Diagnosis from Evidence

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Problem Description

A user reports that the network is slow. A speed-test page returns one number: download throughput. That number does not say whether the default gateway answered, whether a name resolved, whether packets were dropped, or whether delay rose only when the link was loaded. RFC 2330 treats a measurement as usable only when the metric, the path, and the conditions are stated [1].

Key issues:

- Speed tests report throughput. They do not isolate gateway, DNS, loss, or congestion.
- College labs and small teams have no SIEM. The person who notices the fault has to act on it.
- Tools that invent a cause — cut cable, ISP core, Wi-Fi interference — produce an action the measurements cannot support.
- When the four checks do not isolate one cause, the correct output is Inconclusive, not a guessed fault.

Background

Last-mile faults in a campus or a small office are usually local. The gateway may not answer, the resolver may fail while the gateway is fine, the path may drop packets, or delay may jump only under load, which is congestion rather than loss. These four are separable with ordinary probes, and they are the faults a student or a small IT team can act on [1].

The proposed optimization is not a faster speed test. It is a rule that maps the four measurements to one outcome, and a recovery path that changes nothing until the user allows it and the retest confirms the change. Claims that cannot be proved stay outside the system [1].

Literature & RFC Review

RFC 2330 defines the IPPM framework: a metric is meaningful only with a stated path and stated conditions [1]. RFC 2681 defines a round-trip delay metric, and RFC 7680 defines a one-way loss metric [2]. Together they justify measuring delay and loss separately, and refusing a diagnosis when either is missing.

Consumer speed tests measure bulk transfer only. They do not report gateway reachability, resolver success, or loaded versus idle delay. ProbeSpeak uses these RFC metrics as the only inputs to the rule engine [2].

Proposed Solution

ProbeSpeak is a browser diagnosis page and a local recovery agent. The page runs four checks: gateway reachability, DNS resolution, packet loss, and loaded latency. The numbers go to a deterministic rule engine. The engine is not a language model.

- Gateway did not answer → gateway_unreachable
- Gateway answered, DNS failed → dns_failure
- Path loss is high → packet_loss
- Loaded RTT much higher than idle RTT → congestion
- All four inside thresholds → healthy
- Missing or contradictory evidence → inconclusive

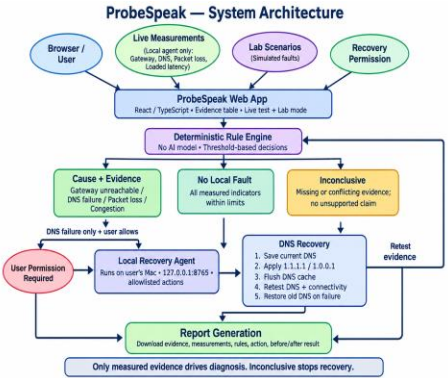
The agent listens only on 127.0.0.1:8765 and accepts a closed list of actions. For DNS it verifies the failure, waits for permission, saves the current servers, applies 1.1.1.1 and 1.0.0.1, flushes the cache, and retests. The change is kept only if the retest passes. Resolved is shown only when that retest passes.

Optimization Proposed

Scenario	Before (Traditional / Current Approach)	After ProbeSpeak (Propesed Solution)	Improvement / Benefit
Slow network	One throughput number	Four checks (gateway, DNS, packet loss, loaded latency) and the rule that fired	Accurate and clear root cause with evidence instead of a single number
DNS failure	Guessed ISP fault	Verified locally, fixed only with permission, retested	Correct diagnosis and safe, permission-based recovery
Unclear evidence	An invented cause	Inconclusive. No recovery offered	No false claims. Only act when evidence clearly matches a cause
Demo for a judge	Break a real network	Lab mode injects the same faults through the same engine	Safe, repeatable and consistent demo without affecting real networks
Agent status	Offline or connected, mixed	Separate states: not installed, starting, connected, unavailable, permission blocked	Clear agent status for better user guidance and troubleshooting

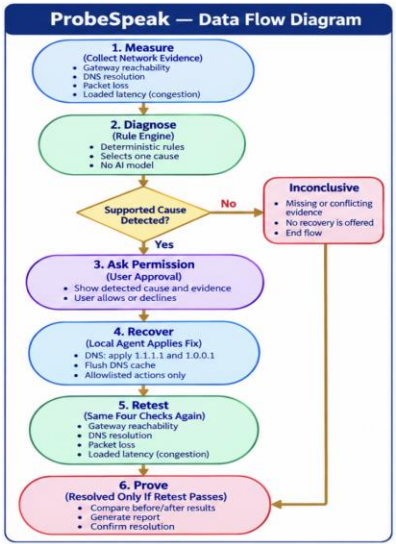
Solution Architecture & Design

The browser measures four signals. The rule engine selects one cause or Inconclusive. Only a supported DNS failure, and only after permission, reaches the local agent. The report records the evidence either way. No unsupported claim is made.

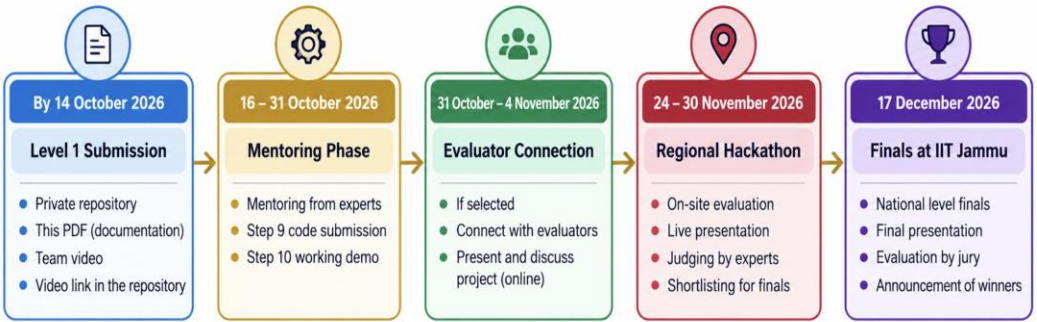


Data Flow Diagram

Measure → Diagnose → Ask permission → Recover → Retest → Prove.
Inconclusive endParticipations the flow. Connected is shown only after the health check succeeds. Resolved is shown only when the retest passes.



Deployment Timeline



Bibliography I

- [1] V. Paxson, G. Almes, J. Mahdavi, and M. Mathis. “Framework for IP Performance Metrics.” RFC 2330. IETF, May 1998.
- [2] G. Almes, S. Kalidindi, and M. Zekauskas. “A Round-trip Delay Metric for IPPM.” RFC 2681. IETF, September 1999.
- [3] G. Almes, S. Kalidindi, M. Zekauskas, and A. Morton. “A One-Way Loss Metric for IP Performance Metrics.” RFC 7680. IETF, January 2016.
- [4] AIORI. “AIORI-3 Online Hackathon Guidelines.” portal.aiori.in/aiori-3/hackathon-guideline. 2026.



Thank You!